



Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Tree Instability and the Bootstrap

WEEK 1 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

One tree, many possible trees

Week 1: Bagging

Average many trees fitted to bootstrap samples

Week 2: Boosting

Trees one after another, each fixing the last

Week 3: Tuning, rare events

Honest tuning, features, imbalance, SHAP

Today's question

If the training rows had been slightly different, how different would the tree be?

One set of rows, many plausible others: draw rows at random, with replacement, as many as there are.

A tree is unstable. An average of many unstable trees is not. That is week 1.

Where Course 4 left the tree

0.7190

logistic regression

0.6161

default tree (day 9)

0.7627

pruned tree (day 10)

Validation ROC AUC, Course 4's 18,000 training rows fit, its 6,000 validation rows report

The default tree

2,750 leaves, 38 questions deep. Training ROC AUC 0.9997: its rows, memorised.

The pruned tree

Class weights, ccp_alpha 0.001 by five folds: 12 leaves, 6 deep. Its output is the model's score.

These three numbers are the bar every Course 5 ensemble on the card file is judged against.

One rule for the whole course: the test rows stay closed

18,000 training rows

6,000 validation

6,000 test

fit; choose by cross-validation

report only

closed

Why closed

Course 4 opened them once, on day 10. Scored again after their numbers were seen, they would become rows the course chooses against.

How it is enforced

`course4_card_rows` rebuilds Course 4's split and never returns a closed row. A test recomputes the closed labels and asserts none comes back.

Validation rows were used in Course 4 to choose, never to fit. Course 5 chooses nothing on them.

The bootstrap: draw with replacement

18,000 draws from 18,000 rows, each row put back after it is drawn. 11,383 distinct rows made it in (63.24 percent); 6,617 never did.

Times drawn	Rows
0	6,617
1	6,662
2	3,236
3	1,150
4	268

Why about 36.8 percent are left out

$$(1 - 1/n)^n \rightarrow e^{-1} \text{ as } n \text{ grows}$$

At $n = 18,000$: 0.367869. e to the power -1 : 0.367879. Each draw misses a row with chance $1 - 1/n$, and the n draws are independent.

A bootstrap sample is the training rows with a random weight on each row.

Ten samples, ten full trees

The first question, in all ten trees

PAY_0 ≤ 1.5

The same as Course 4's default tree. No reshuffle of the rows moves it.

1,832 to 1,921

leaves

35 to 44

questions deep

0.6015 to 0.6219

validation ROC AUC

Below the first few questions, each tree splits on the rows its sample drew twice or left out.

One account, ten outputs

97.7%

outputs exactly 0 or 1

0.2858

average spread across ten

73.2%

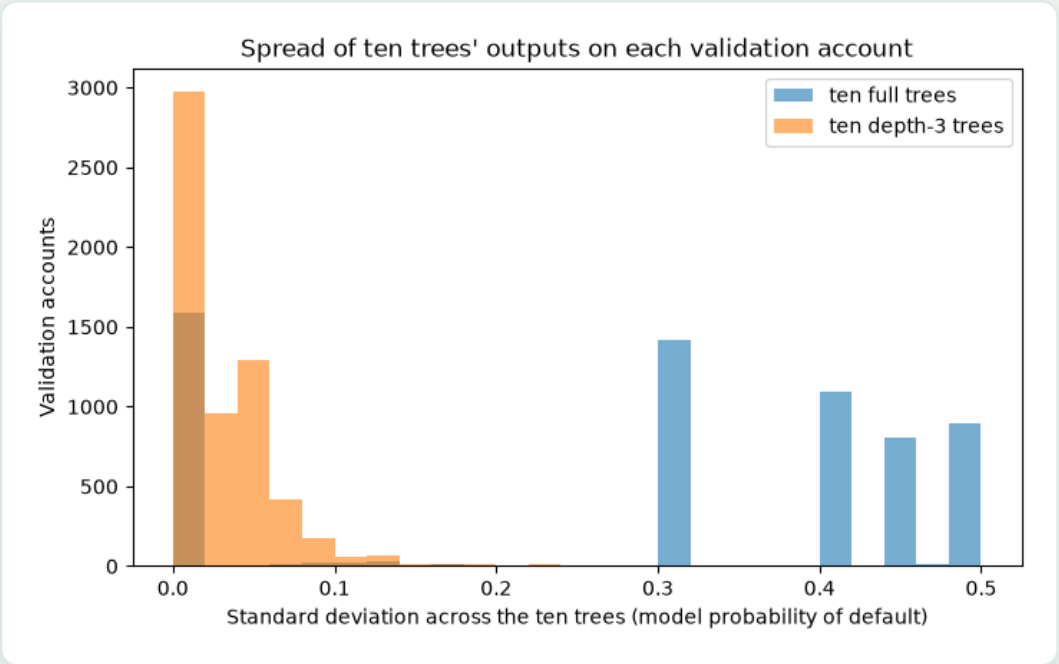
accounts whose flag changes

On 4,395 of 6,000 validation accounts, at least one tree flags the account at 0.5 and at least one does not.

Whether one full tree flags an account depends on which rows it happened to see.

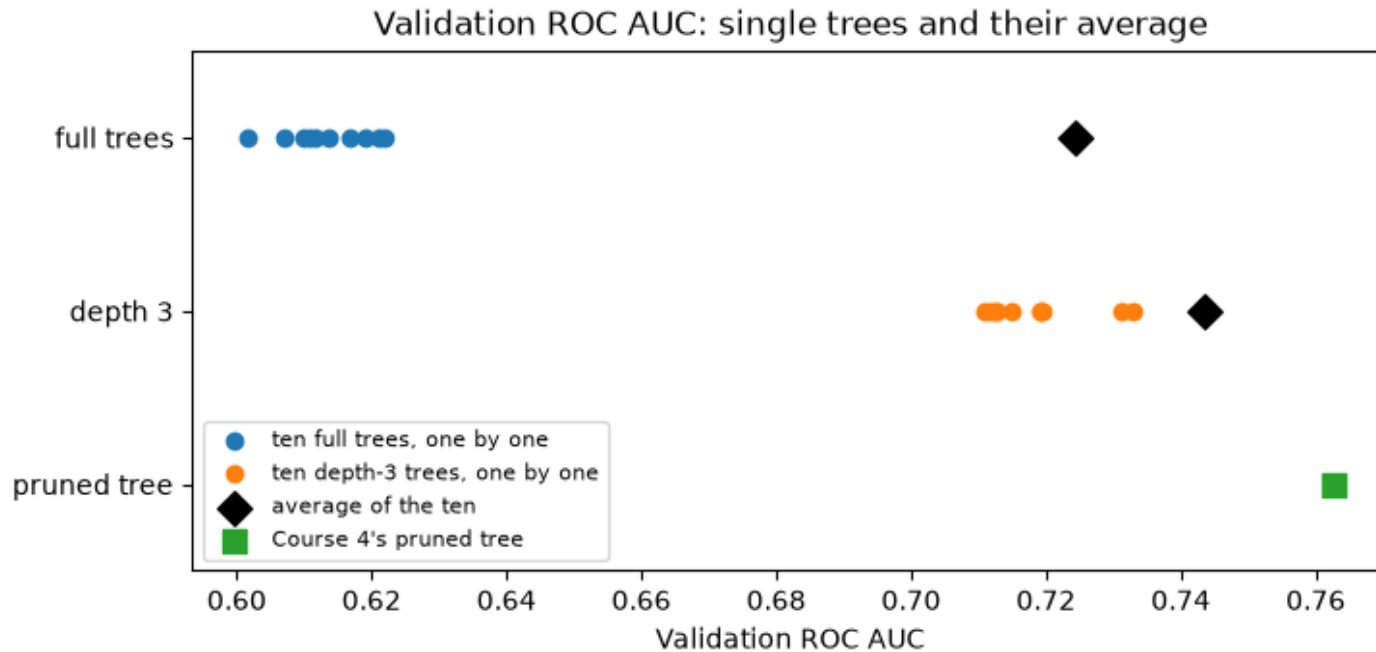
Depth-3 trees: steadier, and weaker

Ten trees	Full	Depth 3
Validation ROC AUC	0.6015 to 0.6219	0.7109 to 0.7330
Average spread	0.2858	0.0306
Flag changes	73.2%	4.3%



A depth limit buys steadiness by giving up detail: still below the pruned tree's 0.7627.

Averaging the ten full trees



0.7243

average of ten full
trees

0.6219

best single full tree

Depth 3, averaged

0.7436, against their best single tree's 0.7330:
they already agree, so little cancels.

Averaging pays where the members disagree.

Where the name comes from

Leo Breiman, August 1996

"Bagging predictors"

Machine Learning, volume 24, issue 2, pages 123 to 140

The name this week uses: fit the same model to many bootstrap samples, and average.

Source, read 5 October 2026: the Crossref record of the paper (DOI 10.1007/BF00058655). Bibliographic facts only; the paper itself was not read for this course.

The out-of-bag rows

The rows a sample never drew. The tree fitted on it never saw them, so they can score it without touching the validation rows.

6,617

rows tree 1 never saw

0.6125

ROC AUC on them

0.6219

ROC AUC on validation

```
left_out = ensemblekit.out_of_bag(n_rows, samples[0])
```

Day 2 scores a whole bagged model this way. 7 tests pass at the end of today.

Excel side by side

Excel side by side (tested in Excel)

`= (1 - 1/n)^n`

`=FREQUENCY(A2:A18001,SEQUENCE(18000,1,0))`

`=COUNTIF($C$2:$C$18001,k)`

`=ROWS(A2:A18001) - COUNTA(UNIQUE(A2:A18001))`

`=STDEV.P(B2:K2), then =AVERAGE(L2:L6001)`

`=AVERAGE(B2:K2)`

`=COUNTIFS(N2:N6001,">0",N2:N6001,"<10")/ROWS(N2:N6001)`

0.367869 at n = 18,000

one count per position, in C2

rows drawn k times: 6,617, 6,662, 3,236

6,617 never drawn

average spread 0.2858

the ten-tree average

flag changes 0.7325

No seeded bootstrap and no tree in Excel. RAND, RANDBETWEEN, RANDARRAY take no seed and recalculate: the positions come from Python.

Save the day: the experiment log

1. Init

The work folder, local settings,
the guard

2. .gitignore

__pycache__/, .pytest_cache/,
*.xlsx

3. Code

mlkit.py with ensemblekit and
its tests

4. Results

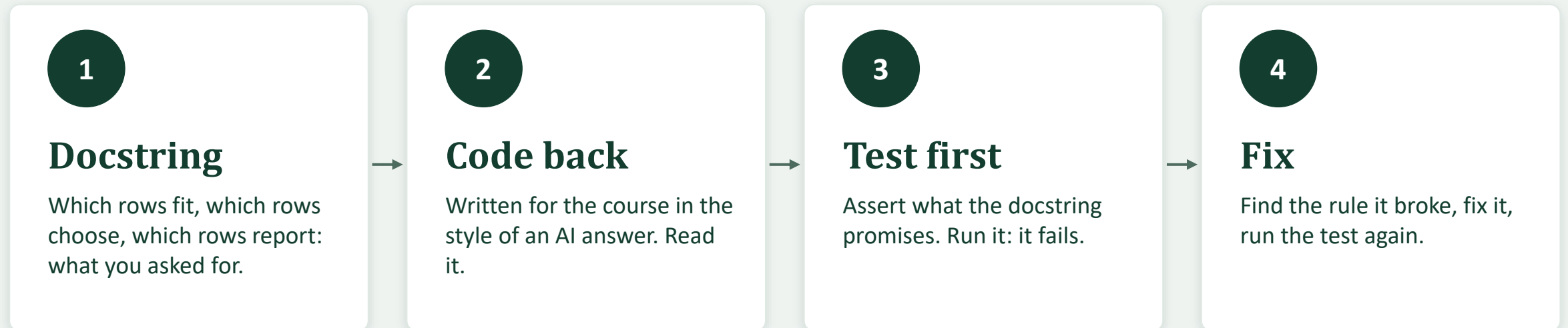
results.csv, 4 decimals

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
course4_default_tree,validation,0.6161,0.2917,0.5000,0.7278,0.3888,0.4032,0.3959
course4_pruned_tree,validation,0.7627,0.4954,0.5000,0.7683,0.4805,0.5840,0.5272
average_of_ten_full_trees,validation,0.7243,0.4453,0.5000,0.7907,0.5349,0.4099,0.4642
```

Threshold 0.5 today; from day 5, out-of-fold. Accuracy beside the all-zero baseline, 0.7788, never to choose.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: the bootstrap is drawn with `rng.permutation`: every row once, no row out of bag.

Code that runs is not the same as code that is right.

What you can do now

- Rebuild Course 4's rows; keep its test rows closed
- Draw a bootstrap sample; about 36.8 percent left out
- A full tree's first question is stable; its leaves are not
- A depth limit buys steadiness and costs detail
- Averaging pays where the members disagree

Week 1, Day 2 Bagging with scikit-learn, and the out-of-bag score